Hands-On-Activity No. 5.1

Hands-on Activity 5.1: Multidimensional Arrays

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed: September 8, 2026**
Section: CPE11S2	**Date Submitted: September 8, 2026**
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1. Objective(s)

This activity aims to develop programming skills using structures.

2. Intended Learning Outcomes

2.1 Create a program using multidimensional arrays.

2.2 Implement multidimensional arrays

3. Materials and Equipment

Internet connection
Personal Computer

4. Output

1. Write a program that creates a multiplication table using multidimensional array.

Answer:

```
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
   const int Rows = 10;
   const int Cols = 10;



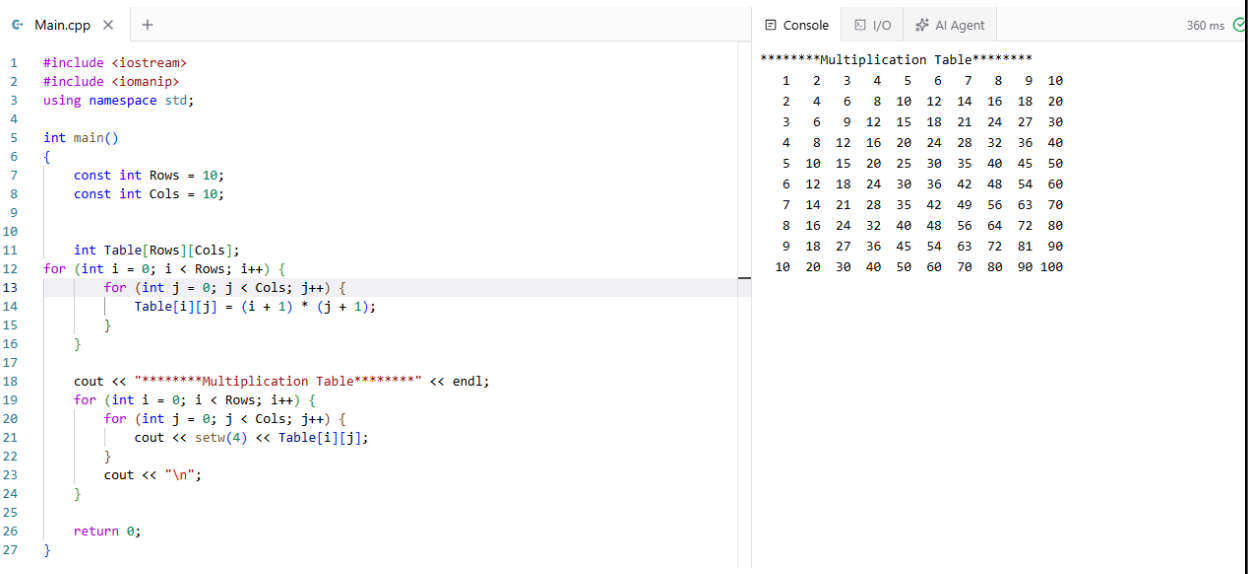
   int Table[Rows][Cols];
for (int i = 0; i < Rows; i++) {
      for (int j = 0; j < Cols; j++) {
         Table[i][j] = (i + 1) * (j + 1);
      }
   }

   cout << "*********Multiplication Table*********" << endl;
   for (int i = 0; i < Rows; i++) {
      for (int j = 0; j < Cols; j++) {
         cout << setw(4) << Table[i][j];
```

```
            }
            cout << "\n";
        }

        return 0;
    }
```

ScreenShot:



2.Write a program that creates a board with a tic-tac-toe moves.

Answer:

```
#include <iostream>

using namespace std;

int main() {
  char board[3][3] = {
    {'1', '2', '3'},
    {'4', '5', '6'},
    {'7', '8', '9'}
  };

  char player = 'X';
  int turn = 0;
  bool gameover = false;
```

```
cout << "Tic Tac Toe Game" << endl;

while (gameover == false) {

    // Print the board
    cout << endl;
    cout << " " << board[0][0] << " | " << board[0][1] << " | " << board[0][2] << endl;
    cout << "---|---|---" << endl;
    cout << " " << board[1][0] << " | " << board[1][1] << " | " << board[1][2] << endl;
    cout << "---|---|---" << endl;
    cout << " " << board[2][0] << " | " << board[2][1] << " | " << board[2][2] << endl;
    cout << endl;

    // Get player input
    int choice;
    cout << "Player " << player << " turn. Enter number (1-9): ";
    cin >> choice;

    if (choice < 1 || choice > 9) {
        cout << "Invalid spot! Try again." << endl;
        continue;
    }

    // Find matching position using for loops
    int row = -1;
    int col = -1;
    int currentNum = 1;

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            if (currentNum == choice) {
                row = i;
                col = j;
            }
            currentNum = currentNum + 1;
        }
```

```
    }

    // Check if space is taken
    if (board[row][col] == 'X' || board[row][col] == 'O') {
        cout << "That spot is already taken!" << endl;
        continue;
    }

    board[row][col] = player;
    turn = turn + 1;

    // Check rows
    for (int i = 0; i < 3; i++) {
        if (board[i][0] == player && board[i][1] == player && board[i][2] == player) {
            gameover = true;
        }
    }

    // Check columns
    for (int j = 0; j < 3; j++) {
        if (board[0][j] == player && board[1][j] == player && board[2][j] == player) {
            gameover = true;
        }
    }

    // Check diagonals
    if (board[0][0] == player && board[1][1] == player && board[2][2] == player) gameover = true;
    if (board[0][2] == player && board[1][1] == player && board[2][0] == player) gameover = true;

    if (gameover == true) {
        // Print final board
        cout << endl;
        cout << " " << board[0][0] << " | " << board[0][1] << " | " << board[0][2] << endl;
        cout << "---|---|---" << endl;
        cout << " " << board[1][0] << " | " << board[1][1] << " | " << board[1][2] << endl;
        cout << "---|---|---" << endl;
        cout << " " << board[2][0] << " | " << board[2][1] << " | " << board[2][2] << endl;
```

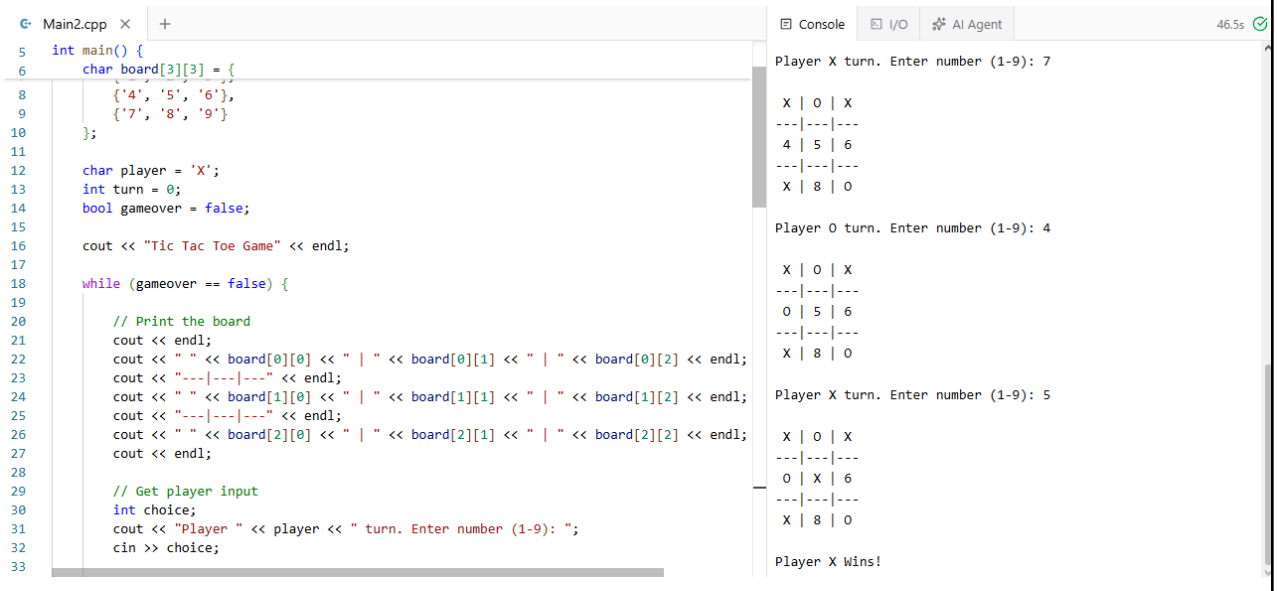
```
        cout << "\nPlayer " << player << " Wins!" << endl;
        break;
    }

    if (turn == 9) {
        cout << "\nIt's a draw!" << endl;
        break;
    }

    // Switch player
    if (player == 'X') {
        player = 'O';
    } else {
        player = 'X';
    }
  }

  return 0;
}
```

ScreenShot:



5. Conclusion

Learning more about arrays and structures for better understanding and when to use.

6. Rubric

SO7